Evaluation of Oregano (*Lippia berlandieri*) Essential Oil and Entomopathogenic Fungi for *Varroa destructor* Control in Colonies of Honey Bee, *Apis mellifera*

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Evaluation of Oregano (*Lippia berlandieri*) Essential Oil and Entomopathogenic Fungi for *Varroa destructor*[1] Control in Colonies of Honey Bee, *Apis mellifera*[2]

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Abstract. Potential of organic compounds to control *Varroa destructor* Anderson & Trueman, on honey bee, *Apis mellifera* L. colonies was evaluated at Cuauhtémoc, Chihuahua, México (28°26'53"N, 106°49'46"W). Treatments were 1.16, 1.33, and 1.50 ml of oregano pure oil (carvacrol 60%); a mixture of oregano oil (7, 8, and 9% carvacrol) with canola oil (v/v); and as a biological control agent, native strains of *Beauveria bassiana* and *Metarhizium anisopliae* (10^8 CFU/g). The efficacy of organic compounds and biological agents was compared with that of amitraz (1%) to eliminate mites from infested colonies of honey bees. In two assays, amitraz was significantly ($P < 0.01$) the most effective acaricide, with a mean efficacy of 97% in eliminating *V. destructor* from infested honey bee colonies. However, the most effective organic compound after amitraz in killing *V. destructor* was 1.16 and 1.5 ml of oregano pure oil, with efficacies of 57 to 74% ($P < 0.01$). Results suggested that oregano pure oil should be considered a viable alternative to synthetic acaricides for controlling *V. destructor* in honey bee colonies. The amount of carvacrol in honey produced by bees treated with oregano essential oil did not exceed the taste detection threshold of 0.1 ppm, and therefore, did not pose a risk of taste modification of natural honey.

Resumen. Se evaluó el potencial de diferentes compuestos orgánicos para el control de *Varroa destructor* en colonias de abejas *Apis mellifera* L., localizadas en Cuauhtémoc, Chihuahua, México (28°26'53"N, 106°49'46"O). Los tratamientos consistieron de 1.16, 1.33, y 1.5 ml de aceite de orégano puro (carvacrol 60%), la combinación de aceite de orégano (7, 8, y 9% carvacrol) con aceite de canola (v/v), y como agente de control biológico, cepas nativas de *Beauveria bassiana* y *Metarhizium anisopliae* (10^8 UFC/g). La eficacia de los compuestos orgánicos y agentes biológicos se comparó con la de amitraz (1%) para eliminar los ácaros de las colonias de abejas infestadas. En los dos ensayos, el amitraz fue significativamente ($P < 0.01$) el acaricida más eficiente con un 97% de eficiencia para eliminar *V. destructor* de las colonias de abejas infestadas. Mientras que, el

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compuesto orgánico más eficiente después de amitraz en la erradicación de *V. destructor* fue el aceite de orégano puro en dosis de 1.16 y 1.5 ml con eficacias que oscilaron desde el 57 al 74% ($P < 0.01$). Estos resultados sugieren que el aceite de orégano puro podría ser considerado como una alternativa viable al uso de acaricidas sintéticos para el control de *V. destructor* en colonias de abejas. Finalmente, el contenido de Carvacrol en la miel tratada con aceite esencial de orégano no superó el umbral de detección del sabor de 0.1 ppm, por lo tanto, no ocasiona un riesgo de modificación del sabor natural de la miel.

Introduction

Varroa destructor Anderson & Trueman is a parasite threatening the survival of the honey bee, *Apis mellifera* L. (Sammataro et al. 2000). In Japan, North America, Europe, and the Middle East, the presence of the parasite has been confirmed and loss of honey bee colonies reported (Neumann and Carreck 2010). Honey bees in South America and Africa better tolerate *V. destructor,* so economic damage is not substantial (Imdorf et al. 1999). *V. destructor* vectors disease (Capinera 2008, Le Conte et al. 2010) by injecting virus into the haemolymph of brood and adult honey bees that later weaken and have malformed bodies and wings (Genersch 2010, Rosenkranz et al. 2010, van Engelsdorp and Meixner 2010), and the colony collapses in 6 to 24 months (Le Conte et al. 2010).

Currently, the most effective way to control *V. destructor* is use of synthetic chemicals (Meikle et al. 2012) such as amitraz, bromopropylate, cymiazole, and flumethrin. However, synthetic chemicals might pose risk to production of honey and beeswax (Kanga et al. 2006). The negative effect of controlling *V. destructor* with synthetic chemicals has started the search for new and less toxic ways to control the parasite. Several essential oils such as camphor, clove, eucalyptus, lemongrass, oregano, and wintergreen with acaricidal activity have been used to control the parasite (Imdorf et al. 1999). The acaricidal effect of essential oils is attributed to monoterpenes (Tripathi et al. 2009). The distribution of monoterpenes in essential oils is an important factor for their effectiveness against *V. destructor* (Ghasemi et al. 2011). Adamczyk et al. (2005) thought essential oils might negatively impact the taste and odor of honey when aromatic compounds exceed the taste threshold of 1.1 $mg\text{-}kg^{-1}$ (Colin 1997).

In addition to essential oils, entomopathogenic fungi such as *Beauveria bassiana* (Balsamo) Vuillemin (Meikle et al. 2007, 2008), *Metarhizium anisopliae* (Metschnikoff) Sorokin (Kanga et al. 2003), and *Hirsutella thompsonii* Fisher (Hypocreales) are effective antagonists of *V. destructor*. Combined use of essential oils and entomopathogenic fungi might be a viable alternative to control the parasite. However, efficacy of essential oils and biological agents might depend on factors such as weather, dose, and concentration.

Several authors suggested the factors mentioned might cause variability in the efficacy of different treatments assessed against *V. destructor* in the field (Maggi et al. 2011). Therefore, the aim of this study was to evaluate the efficacy of oregano essential oil and two entomopathogenic fungi for control of *Varroa destructor* in the climatic conditions at Cuauhtémoc, Chihuahua, Mexico.

Materials and Methods

Efficacy of oregano essential oil and biological control agents were evaluated against *V. destructor* in experiments in November 2011 and October 2012. Both experiments were in an apiary at the Centro de Investigación en Alimentación y Desarrollo A.C. (28°26' 53'' N; 106°49' 46'' W, and 2,038 m above sea level, in temperate climatic zone) at Cuauhtémoc, Chihuahua, Mexico. Thirty naturally infested Langstroth beehives were used for each experiment. The 10 treatments were randomly distributed among 30 hives as described in Table 1.

Absorbent cotton towels impregnated with oregano essential oil and amitraz 24 hours before the assays were placed on top of the combs. A salt shaker was used to manually dust *B. bassiana* and *M. anisopliae* onto the honeycomb plates (Kanga et al. 2003).

Diluted essential oil of oregano (1:100 in hexane) was characterized by gas chromatography mass spectrometry using a Varian Saturn 2100D system equipped with NIST 2008 library and fused silica capillary column HP-5MS (30 m × 0.25 mm; 0.25 µm film thickness, Hewlett Packard, Palo Alto, Ca.). The gas chromatography oven temperature was increased from 55 to 65°C at the rate of 1°C per minute, held 3 minutes, and increased to 290°C at the rate of 10°C per minute. Helium gas at a flow rate of 1 ml per minute was the carrier. Electron ionization mass spectra (70 eV) were recorded by scanning from 30 to 500 m/z. Injector and transfer line temperatures were maintained at 250°C. One microliter of each sample was injected in a split mode (50:1). Composition of oregano essential oil was reported as the relative percentage of peak areas of the main components previously reported to have acaricidal activity. The main components were identified by mass spectra correlation using the NIST library and matching the retention times of high purity standards (Sigma-Aldrich, Toluca, Mexico).

B. bassiana and *M. anisopliae* strains were isolated from insects in the genera *Melanoplus* sp. (Orthoptera: Acrididae) and *Phyllophaga* sp. (Coleoptera: Scarabaeidae), respectively, from Cuauhtémoc, Chihuahua, Mexico. Both entomopathogenic fungi were purified and identified according to morphological characters. Further characterization was by PCR using a ZR Fungal/Bacterial DNA MiniPrep kit (Zymo Research) to extract DNA, and preformed primers by Hegedus and Khachatourians (1996) and Schneider et al. (2011) to confirm the presence of *B. bassiana* and *M. anisopliae*, respectively. Entomopathogenic fungal strains were

Table 1. Treatments Evaluated against *Varroa destructor* of Honey Bees in November 2011 and October 2012

Treatment	Description
Synthetic acaricide	10 ml of amitraz (1%) on cotton towel (9 × 27 cm)
B. bassiana	15 g of 10^8 CFU/g of dry conidia
M. anisopliae	15 g of a 10^8 CFU/g of dry conidia
Oregano essential oil	1.16 ml (carvacrol 60%) on cotton towel (8 × 9 cm)
Oregano essential oil	1.33 ml (carvacrol 60%) on cotton towel (8 × 9 cm)
Oregano essential oil	1.5 ml (carvacrol 60%), on cotton towel (8 × 9 cm)
Canola & oregano oil mixture	8.84-1.16 ml (v/v), on cotton towel (9 × 27 cm)
Canola & oregano oil mixture	8.84-1.33 ml (v/v), on cotton towel (9 × 27cm)
Canola & oregano oil mixture	8.84-1.5 ml (v/v), on cotton towel (9 × 27 cm)
Check	No application

characterized by solid-state fermentation, with slight modification. Two hundred fifty milliliters of sterilized mineral salt media (10 g sucrose, 5 g yeast extract, 0.2 g K_2HPO_4, 0.2 g $MgSO_4H_2O$, and 0.2 g $MnSO_4$) were inoculated with purified strain spores and kept at 28 ± 1°C with constant stirring at 180 rpm for 7 days to obtain spore inoculum. The substrate for culturing fungi was 250 g of damp, sterilized rice (121°C, 15 minutes, 15 psi) in rectangular polypropylene trays (25 × 18 × 4.5 cm) inoculated with 50 ml of spore inoculum and incubated in an environmental growth chamber (27 ± 1°C, 21 days). The content of the trays was pulverized in a mortar to obtain dried biological agent. The number of colony-forming units per gram of dried powder was determined in a Neubauer hemocytometer (BOECO, Germany). The number of colony-forming units per milliliter was assessed after plating 100 µl of serial dilutions of pulverized biological agents onto potato dextrose agar and incubating at 32°C for 96 hours.

Mite infestation percentage in the selected beehives was assessed by the ethanol wash method (De Jong et al. 1982) before beginning the experiment. Two hundred adult worker bees were placed into a 4-liter glass jar containing 100 ml of diluted ethanol (70%). The jar was shaken mechanically for 3 to 5 minutes (100 rpm) and the ethanol solution discarded through a mesh. The number of mites on the sieve divided by the number of bees was the infestation percentage.

Efficacy of treatments (100 times the number killed by treatment divided by the total number of *V. destructor* in the beehive) was evaluated based on number of dead mites after each treatment. To assess the number of dead mites, at the beginning of the experiment a white piece of paper coated with petroleum jelly was placed on the bottom of the beehive to catch dead mites. Stuck-dead *V. destructor* were counted every 3 or 4 days until the end of the experiments totaling five samples per treatment. Mite catchers were replaced on days 7 and 14. Amitraz (3% v/v) was applied on Day 21 of the experiments to evaluate the number of *V. destructor* remaining after each treatment.

Carvacrol residue in honey was determined by head-space solid phase micro-extraction gas chromatography coupled to mass spectrometry according to Cuevas-Glory et al. (2008), with slight modification. Honey bees (6 g) were transferred to a 10-ml headspace vial and sealed with a PTFE/silicon septum. Honey was conditioned in a heater block at 70°C for 30 minutes and volatiles adsorbed onto an SPME fiber (1 cm 65 µm polydimethylsiloxane/divinyl benzene; Supelco, (Bellefonte, PA.) for an additional 40 minutes. After concentration, volatiles were thermally desorbed (250°C) for 5 minutes into a gas chromatograph injection port. A Varian 3800 gas chromatograph equipped with Varian Saturn 2100D mass spectrometer detector was used for analysis of volatile flavor compounds. The gas chromatograph mass spectrometer was equipped with a fused-silica capillary column (DB-WAX, 30 m x 0.25 mm x 0.25 µm film thickness, Agilent Technologies). The oven initially was kept at 50°C for 4 minutes, increased at a rate of 10°C per minute to 200°C, kept for 3 minutes, and increased to a final temperature of 230°C at a rate of 20°C per minute. Helium was the carrier with a flow of 1 ml per minute. The mass spectrometer operated from 30 to 500 m/z in electron ionization mode (70 eV). Carvacrol in honey was identified using a standard (Sigma-Aldrich, Toluca, Mexico.) and confirmed by mass spectra correlation using the NIST 2008 library.

The experimental design was completely randomized with 10 treatments and three replications, with each hive representing an experimental unit. Data on percentage of efficacy was arcsine transformed and subjected to ANOVA by orthogonal contrast with SAS System software (SAS Institute, Cary, NC).

Results

Chromatographic analysis of oregano essential oil indicated phenolic and monoterpene hydrocarbons (Table 2). Percentages ranged from 59.3% carvacrol to 5.2% terpene.

According to macro- and microscopic morphological observations of the fungi isolated, the putative strain of *B. bassiana* had hyaline conidiophores with zigzag branching, globose to ellipsoidal conidia grouped in clusters, and white mycelium with cottony appearance. *M. anisopliae* had hyaline conidiophores, conidia grouped in columns with cylindrical form, slight pigmentation, and yellow to olive-green mycelia. Molecular characterization showed PCR products of 524 and 335 bp for *B. bassiana* (Fig. 1) and *M. anisopliae* (Fig. 2), respectively. The final concentration for both fungi was 10^8 CFU/g formulated, and the density of CFU/g was 100%.

Table 2. Main Components of Oregano Essential Oil Expressed as Percentage of Chromatographic Area >0.1%

Compound	Percent
Carvacrol	59.29
ρ-Cymeno	28.65
Thymol	6.73
γ-Terpenes	5.20

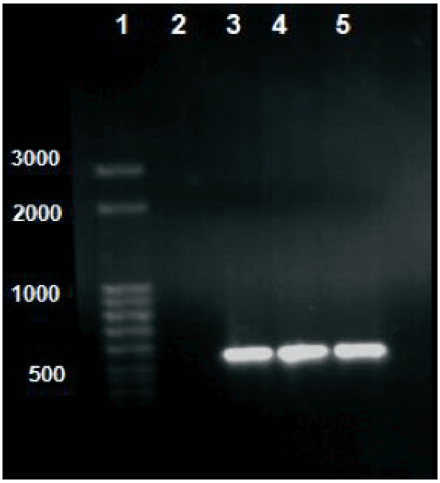

Fig. 1. Ethidium bromide-stained agarose gel showing PCR amplification products of 524 bp obtained from *B. bassiana*. Lanes: 1, molecular weight marker (50 kb); 2, negative check; 3, 4 and 5, *B. bassiana* isolated strains.

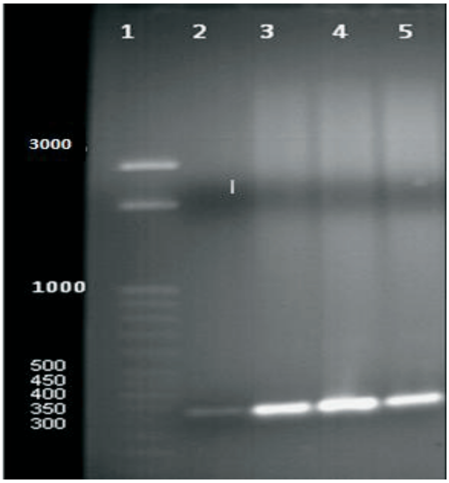

Fig. 2. Ethidium bromide-stained agarose gel showing PCR amplification products of 335 bp obtained from *M. anisopliae*. Lanes: 1, molecular weight marker (50 kb); 2, negative check; 3, 4 and 5, *M. anisopliae* isolated strains.

Infestation of honey bees by *V. destructor* ranged from a minimum of 2.87 and 1.42% to maximum of 18.75 and 17.59% in 2011 and 2012, respectively. The wide ranges of infestation allowed grouping into high, medium, and low, as described by Vandame (2000).

During the first experiment (2011), different treatments killed significantly different percentages of *Varroa* mites (F = 8.96, df = 11, P < 0.0001). The synthetic acaricide killed 98% of mites, followed by 1.16 ml of oregano pure essential oil that killed 74% and 1.5 ml of oregano pure oil (Fig. 3). All other treatments killed significantly fewer mites than the check.

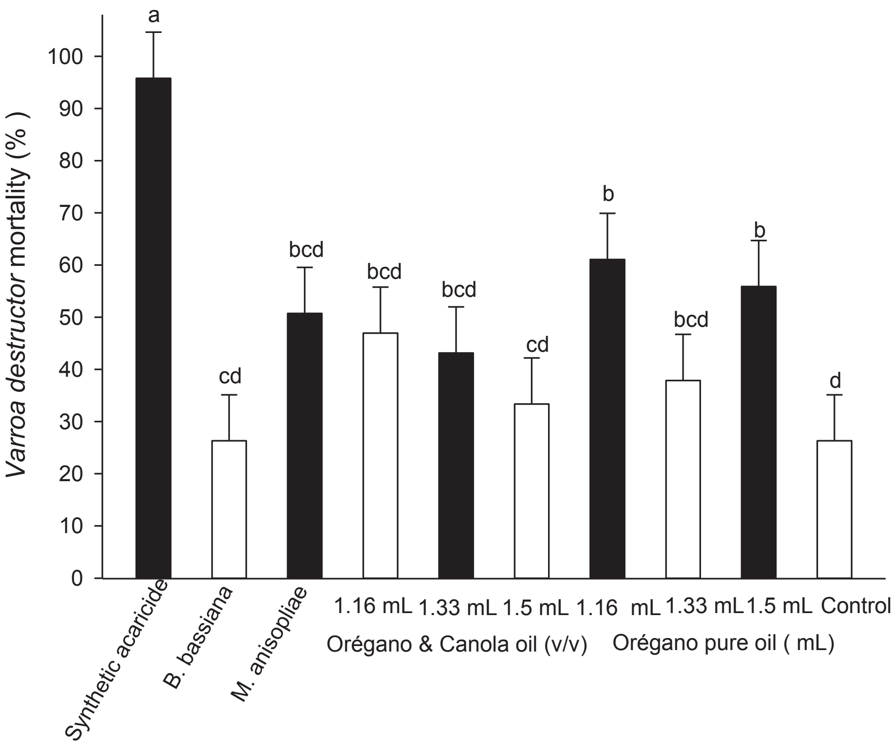

Fig. 3. Mean percentage of *Varroa* dead after treatment in the first experiment (2011). Means with the same letter are not significantly different (P < 0.0001).

In the second experiment (2012), significantly greater efficacy resulted from treatment with synthetic acaricide. Treatments with oregano pure essential oil at any dose, oregano-canola at 1.16 or 1.33 ml, and *M. anisopliae* did not differ significantly from each other. Only oregano pure oil at 1.16 and 1.5 ml and the synthetic acaricide, with 61, 57, and 96% efficacy, respectively, killed significantly more than did the check (F = 8.96, df = 11, P < 0.0001) (Fig. 4). During the first experiment (November 2011), temperature averaged 16°C and relative humidity was 55%, while in the second experiment (October 2012), average temperature was 21°C, with 55% relative humidity.

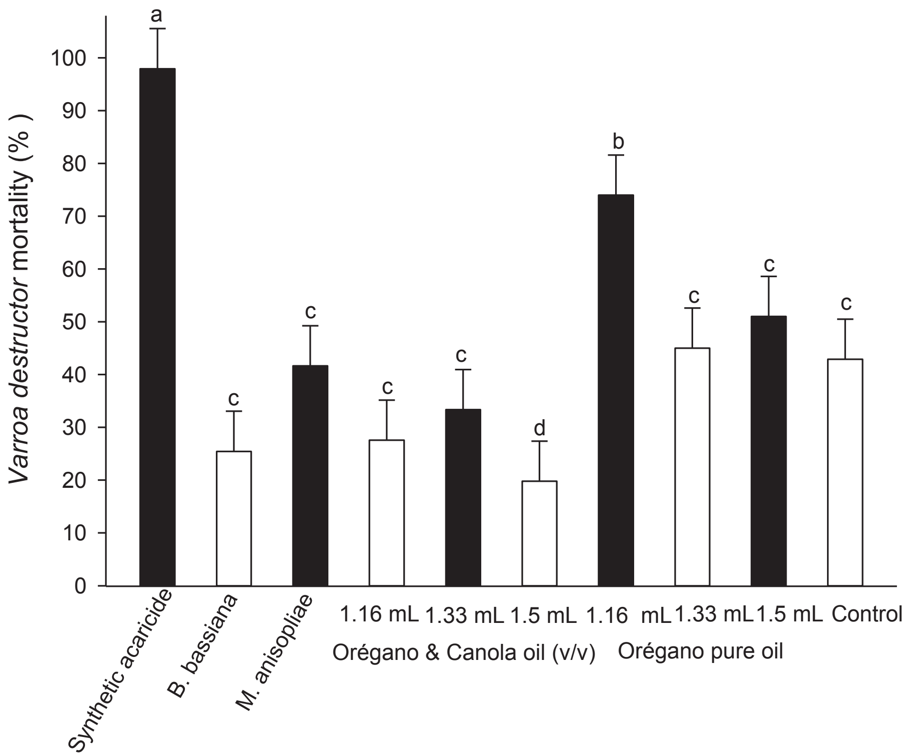

Fig. 4. Mean percentage of *Varroa* dead after treatment in the second experiment (2012). Means with the same letter are not significantly different ($P < 0.0001$).

Chromatographic analysis of honey from beehives treated with oregano essential oil (carvacrol 60%) indicated no treatment posed any risk of changing the taste (Fig. 5). The principal component in the oil did not exceed the limit of the taste threshold of aromatic compounds in honey (1.1 mg/kg).

Discussion

Greatest efficacy (74%) was at the lowest dose of oregano essential oil (1.16 ml), contrary to expectation. This might be because repellent response of bees to doses of 1.5 and 1.33 ml increased circulation of air or temperature in the hive and enabled rapid evaporation of essential oils. Umpiérrez et al. (2011) observed increased temperature of the hive and efficacy of essential oils, which attributed to activity of bees. During the second experiment, significant differences were not observed between the same treatments. Only doses of 1.16 and 1.5 ml, with 61 and 57% efficacy, respectively, were statistically different from the check. This might be caused by temperature effect on release of oils. According to Gouinguene and Trulings (2002), optimal release of volatile compounds in plants is at relative humidity of 60% and temperature between 22 and 27°C. However, Higes et al. (1997) reported problems in the behavior of bees when assessing effect of the commercial acaricide Apilife Var (76% thymol, 16% eucalyptol, 4% menthol, and 4% camphor) that were attributed to the toxicity of eucalyptol, caused by rapid evaporation at temperatures to 27°C. We suggested an average temperature of 21°C might cause rapid evaporation of the oils, which decreased their exposure time and effect. In hives treated with 1.33 and 1.5 ml of oregano essential oil, bees removed the towels where the treatment was applied, but the behavior was not observed in hives treated with 1.16 ml of oregano essential oil, which suggested the lesser dose was tolerated by the bees.

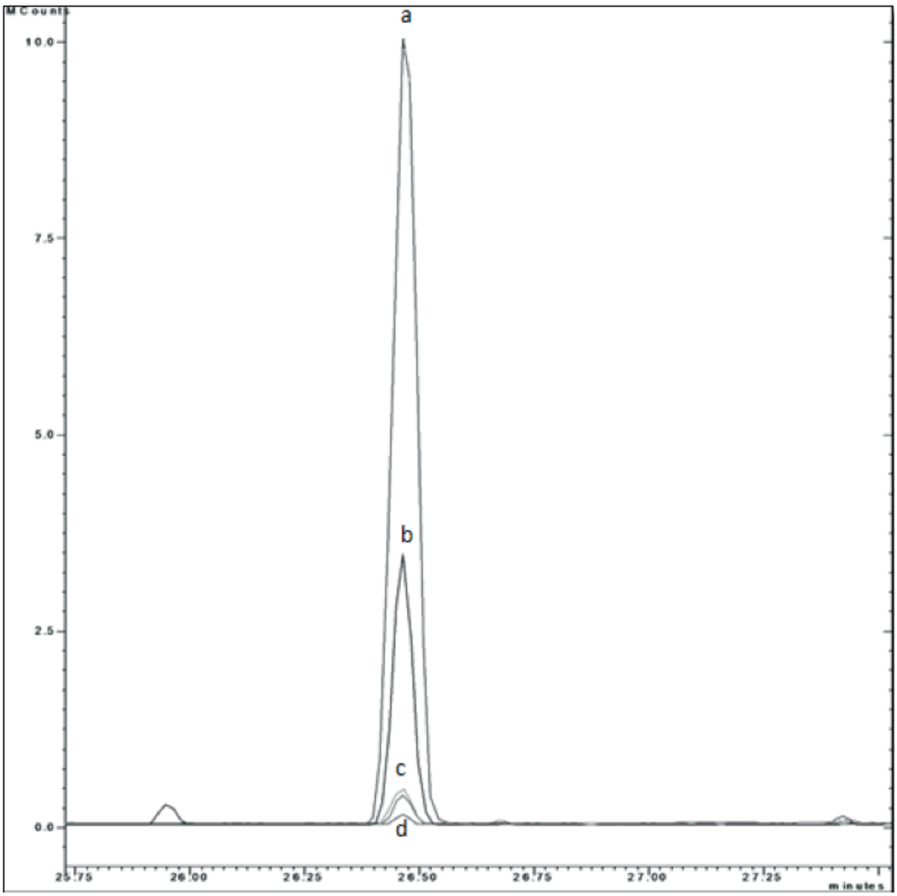

Fig. 5. Chromatogram of carvacrol residues in honey treated with oregano essential oil. Peaks: carvacrol standard 1.1 mg/kg (a), carvacrol in honey oregano treatment 1.5 ml (b), carvacrol in honey oregano treatment 1.33 and 1.16 ml (c), and carvacrol in the honey check.

Efficacies with all doses of oregano-canola oil mixtures were not significantly different from that of the check. This coincided with data by Rice et al. (2002) who evaluated the effect of thymol (1.2%)-canola mixture supplied in a long-release device to control *V. destructor*, finding efficacy of 35% but showing no significant difference from that of the check. Further study of application methods is needed to improve efficacy of the treatments to consider them as a viable alternative.

Meikle et al. (2008) found efficacy of 98% from treatment with 1 g of *B. bassiana* mixed with carnauba wax as a carrier, with a concentration of 3.7×10^{10} CFU/g. Rodríguez et al. (2009) evaluated conidia of *M. anisopliae* applied by dusting a concentration of 5×10^{10} CFU/g and found 98 and 95% efficacy during September and October, respectively. This contrasts with findings of our study because efficacy of *B. bassiana* or *M. anisopliae* was not significantly different from that of the check. In both studies, the concentration was greater than that in our study. Low efficacy might also be attributed to weather, because optimal temperatures for growth of the entomopathogenic fungi range from 25 to 30°C (Zimmermann 2007a), being inhibited at temperatures cooler than 15°C or warmer than 33°C. This suggested that with temperatures of 16 to 21°C during the first and second experiments, optimal temperatures were not reached for development of the fungi. Requirements for both fungi exceeded 90% relative humidity, but studies by Zimmermann (2007ab) reported epizootics at 60 and 70% relative humidity generated in the microclimate on the surface of the integument of insects. In this study, the average relative humidity for both experiments was less than 60%, which indicated humidity might have hindered the efficacy of treatments. Variability of pathogenicity also exists between different strains, as well as intrinsic factors such as the immune response of the host (Schumacher and Poehling 2012).

During the second experiment, bees left two hives days after treatment with 15 g of *M. anisopliae*. Kanga et al. (2003) evaluated the efficacy of the same fungi applied by dusting 46.8 and 96.3 g in a concentration of 10^{10} CFU/g onto a hive and found similar efficacy to synthetic acaricides without significant differences in numbers of bees killed compared to the check. However, they indicated that one hive treated with 46.8 g and two hives treated with 96.3 g swarmed or absconded. The author found that one hive in the check group also absconded, but indicated that other factors might have caused the phenomenon. Further study is necessary to determine the probable cause.

The greatest efficacy both years was by the synthetic acaricide. According to Melathopoulos (1999), efficacy of a treatment in the field must be greater than 70% to be a viable alternative. This suggests that a dose of 1.16 ml of oregano essential oil administered during November could be a viable alternative to control *V. destructor*.

None of the samples treated with oregano essential oil had carvacrol residues in honey greater than the taste threshold (1.1 mg/kg) of aromatic compounds. Adamczyk et al. (2005) who analyzed residues in honey after application of Apilife var (76% thymol, 16% eucalyptol, 4% menthol, and 4% camphor) found only thymol residues in concentrations greater than 1.1 mg/kg. This could be explained because thymol was present in greater proportion in the acaricide, but in the oregano essential oil used in our study, the proportion of the principal component (carvacrol) was only 60%, with which efficient control of *V. destructor* was achieved without exceeding the taste threshold.

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